PROJECT PAPER

TODAGO TAYTAY

Tricycle Fare Calculator & Route Directory

A Flutter-Based Mobile Application

Presented to:
Ace York A. Mangalino

Course / Subject:
IT-E3L

Prepared by:
Matte, Diza D.

Program / Year:
BS in Information Technology – 3

Educational Institution:
Palawan State University-Taytay Campus

Date:
November 2026

TABLE OF CONTENTS

TITLE PAGE i

TABLE OF CONTENTS ii

1. INTRODUCTION 1

2. PROJECT OVERVIEW 2

3. TARGET AUDIENCE 3

4. FEATURES 4

5. FLUTTER TECHNOLOGY 5

6. TESTING AND EVALUATION 6

7. PROJECT TIMELINE AND BUDGET 7

7.1 Estimated Budget 8

8. REFERENCES 9

9. DEVELOPER PROFILE 10

9.1 Technical Skills 10

Project Contribution 10

1. INTRODUCTION

Tricycles are an important mode of short-distance transportation for people living in and visiting Taytay, Palawan. Residents, students, workers, and tourists may need to travel between homes, schools, the public market, local establishments, public offices, and other destinations. However, commuters may not always know which tricycle route to take, where to board, or how much a trip may cost.

To help address these concerns, this project proposes TodaGo Taytay: Tricycle Fare Calculator & Route Directory. The application will organize selected tricycle route information in Taytay and provide a simple way to estimate fares based on the available fare information entered into the system. It is intended as a commuter-information guide. Estimated fares must still be confirmed with the driver or the appropriate local authority because actual fares may change.

The project will use the Flutter framework and Dart programming language. Flutter is suitable because it supports Android and iOS development using a shared codebase, provides reusable interface widgets, and offers Hot Reload to help developers quickly see code changes. These features can support the development of a consistent and responsive mobile application (Flutter, n.d.-a; Flutter, n.d.-b).

2. PROJECT OVERVIEW

TodaGo Taytay is a proposed mobile application focused specifically on tricycle fare estimation and route information within Taytay, Palawan. Its main purpose is to help commuters identify available tricycle routes, choose pickup and drop-off locations, review route details, and view an estimated fare before travelling.

The project has the following objectives:

- Provide an organized directory of selected tricycle routes and commonly used destinations in Taytay, Palawan.
- Allow users to select pickup and drop-off points from available locations.
- Display estimated fare information clearly and remind users to confirm the fare with the driver.
- Present route details to help commuters understand their possible trip.
- Develop a simple mobile interface using Flutter that can support Android and iOS.

3. TARGET AUDIENCE

The primary users of TodaGo Taytay are people who use or need information about tricycle transportation in Taytay, Palawan. These include:

- Residents and daily commuters who regularly travel within the municipality.
- Students who travel to and from schools and other learning institutions.
- Employees and local workers who commute to workplaces and business areas.
- Tourists and visitors who may be unfamiliar with local destinations and tricycle routes.
- Tricycle drivers and operators who may use the directory as a reference for route information, subject to verification.

The application is intended for mobile users, particularly Android users during the initial development and testing stage. Flutter also provides a pathway for supporting iOS devices through the same project codebase.

4. FEATURES

The application will focus on the following core features. The information shown in the directory and fare calculator will depend on verified local data available for the project.

- **Tricycle Fare Calculator:** Allows users to choose a pickup point and destination and view an estimated fare based on the fare data configured in the application.
- **Route Directory:** Lists selected tricycle routes and the destinations or areas covered in Taytay, Palawan.
- **Pickup and Drop-off Selection:** Allows users to identify where their trip begins and where they intend to go.
- **Route Details:** Presents relevant route information, such as route name, pickup point, drop-off point, and other available travel notes.

The application will display fares as estimates and will not represent unverified amounts as official or guaranteed fares.

5. FLUTTER TECHNOLOGY

TodaGo Taytay will be developed using Flutter and Dart. Flutter's widget-based structure will be used to build the screens, buttons, input fields, route lists, and

navigation elements. Reusable widgets can help keep the design consistent across the application.

The planned tools and packages are:

- Flutter SDK and Dart – for building the mobile application and implementing its functions.
- Visual Studio Code or Android Studio – for writing, running, and debugging the application.
- Flutter Material widgets – for common interface elements such as buttons, cards, text fields, and navigation.
- geolocator (optional) – may be considered if the final prototype includes a location-based feature and the required permissions are available.

Only packages needed by the final implemented features will be included. If the project uses a location package, the app will request permission before accessing a device's location.

6. TESTING AND EVALUATION

Testing will focus on the accuracy of the fare-estimation process, the usability of the route directory, and the reliability of the main navigation flow. Flutter's testing tools will be used where appropriate to check individual functions and interface behavior (Flutter, n.d.-c).

- Fare calculation testing – check that the estimated fare changes correctly when different pickup and destination options are selected.
- Route directory testing – confirm that listed routes and destination details appear correctly.
- Pickup and drop-off testing – check that users can select the intended starting point and destination.
- Navigation testing – confirm that users can move between the main screens without errors.
- Usability testing – ask selected users to try the main features and provide feedback about clarity and ease of use.

Route and fare information will be reviewed against available and reliable local references before being presented as project data. Any information that has not been verified will be labeled accordingly.

7. PROJECT TIMELINE AND BUDGET

The project is planned over four weeks, from October 14 to November 18, 2026. The schedule covers project setup, interface design, Flutter development, testing, and final presentation.

Week	**Activities**	**Schedule**
Week 1	Project Setup and UI Design	October 14-21, 2026

Week 2	UI Design and Flutter Development	October 26-November 3, 2026
Week 3	Functionality and Testing	November 4-11, 2026
Week 4	Finalization and Presentation	November 12-18, 2026

7.1 Estimated Budget

The following is a proposed student-project estimate in Philippine pesos. Actual expenses may change depending on available devices, internet usage, printing needs, and the services selected.

Item	**Estimated Cost**	**Notes**
Flutter SDK and Dart	₱0	Open-source tools
VS Code / Android Studio	₱0	Free development tools
Internet / data	₱500–₱1,000	Estimated project use
Testing device	₱0	Assumes an available phone
Printing and documentation	₱300–₱700	Estimated printing costs

Estimated total: ₱800–₱1,700

Cost-saving approach: use free development tools, one Flutter codebase, and an available mobile phone for testing. The budget is an estimate and does not include optional paid services.

8. REFERENCES

Baseflow. (n.d.). Geolocator. pub.dev. https://pub.dev/packages/geolocator

Dart. (n.d.). Dart documentation. https://dart.dev/guides

Flutter. (n.d.-a). Flutter documentation. https://docs.flutter.dev/

Flutter. (n.d.-b). Building user interfaces with Flutter. https://docs.flutter.dev/ui

Flutter. (n.d.-c). Testing Flutter apps. https://docs.flutter.dev/testing

9. DEVELOPER PROFILE

The following profile presents the developer's academic background, project role, and relevant technical skills.



Full Name: Matte, Diza D.

Course/Year/Section: BS in Information Technology -3

Institution: Palawan State University -Taytay Campus

Project Role: UI/UX Designer, Developer, and Documentation

Email: 2024100031t@psu.palawan.edu.ph

Short Biography: I am a third-year Bachelor of Science in Information Technology student at Palawan State University -Taytay Campus. For this project, I contribute to the interface design and planning of TodaGo Taytay, with the goal of making local tricycle fare and route information easier for commuters to understand.

9.1 Technical Skills

Flutter and Dart: Basic mobile app development.

UI/UX Design: Figma and interface design.

Programming: Java, SQL, HTML, CSS, and JavaScript.

Development Tools: VS Code, Android Studio, and Microsoft Word.

Other Skills: Documentation, testing, and troubleshooting.

Project Contribution: Designed the TodaGo Taytay screens and helped prepare the project documentation.